Project Proposal: Analysis & Design Phase

Project: Smart Clinic — an AI assistant that guides patients before they visit the doctor

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This document covers the **Analysis and Design** sections of the **Smart Clinic** proposal. The Analysis section explains the problem, the stakeholders, and whether the project is feasible. The Design section turns that analysis into objectives, a system architecture, a timeline, and a risk plan. It builds on the Project Proposal, the Literature Review, and the Requirements Gathering documents.

1. Project Analysis

The Analysis section diagnoses the current environment, details the problem scope, verifies operational and financial constraints, and assesses initial implementation risks.

1.1 Problem Statement & Needs Assessment

Establishes the gap between the current state and desired performance. Requirements are categorized into felt needs (stakeholder requests) and normative needs (regulatory or industry standards).

SMART CLINIC: PROBLEM STATEMENT & NEEDS

Problem Statement: Patients with chest and respiratory symptoms (cough, fever, shortness of breath) have no simple, reliable way to understand their situation before seeing a doctor. They search the internet and find frightening and contradictory results, they visit the wrong specialty and are referred onward, or they receive an X-ray and a report written in medical language they cannot interpret. The literature confirms the weakness of current tools: an audit of 23 symptom checkers found the correct diagnosis listed first in only about a third of cases and appropriate triage advice in a little over half (Semigran et al., 2015), and later work found large differences in accuracy and safety between apps (Gilbert et al., 2020). Existing tools also work on symptoms alone, are rarely tied to a specialty, and rarely support Arabic phrasing. The result is unnecessary worry, delays in reaching the right doctor, and crowded clinics because of cases that could have been directed correctly from the start.

Desired State: A patient describes their symptoms, uploads a chest X-ray and its report, and within about 10 seconds receives one clear summary in simple language with a suggested specialty and a document to show the doctor.

Needs Assessment:

Type	Need	Source
Felt need	Simple language and a clear next step instead of confusing search results.	Patients
Felt need	The right specialty on the first visit, with a ready summary that saves time in the first examination.	Patients, doctors and clinics
Felt need	Ability to write symptoms in Arabic or Egyptian phrases.	Patients
Felt need	A working, well-documented product delivered on time within the approved scope.	Project Owner (Instructor)

Type	Need	Source
Normative need	The output must be presented as a preliminary guide, never as a diagnosis, with the medical disclaimer on every page (FR-12, C-06).	Medical and ethical practice
Normative need	No permanent storage of patient inputs; only public, anonymised datasets are used (NFR-04, C-02).	Privacy and data-protection principles
Normative need	Respect of dataset licences and proper citation of Kaggle and Open-i.	Dataset providers
Normative need	Urgent symptoms (for example severe shortness of breath or chest pain) must trigger a warning to seek immediate care (FR-13).	Patient safety

1.2 Stakeholder Analysis

Maps individuals and groups affected by the project, detailing their level of influence and core expectations.

SMART CLINIC: STAKEHOLDER MATRIX

Stakeholder Group	Impact / Power	Core Interests & Project Expectations
Project Owner (Instructor)	High / High	A working, well-documented product that matches the approved scope and is delivered on time.
Development team (6 members)	High / High	Clear scope, available data, and enough time and tools to deliver.
Patients (primary users)	Medium / High	Simple language, a clear next step, and the right specialty without confusion.
Doctors and clinics	Medium / Medium	A short, accurate summary that saves time in the first examination, and a clear statement that it is not a diagnosis.
Dataset providers (Kaggle, Open-i)	Low / Low	Respect of licences and proper citation.
Cloud provider (Azure)	Low / Low	Use within free or student limits.

Engagement Approach: Manage the Project Owner and the development team closely; keep patients and doctors informed through wireframe walkthroughs and short feedback sessions; monitor dataset and cloud providers for licence or limit changes.

Conflict Identified Early: Patients want a clear answer, while the Project Owner and doctors need the output to remain a preliminary guide. This is resolved by wording results as “leans towards” and by showing the disclaimer on every page.

1.3 Feasibility Study

Verifies whether the project can realistically be executed within technical, organizational, and economic limits.

SMART CLINIC: FEASIBILITY PROFILE

- **Technical Feasibility:** Feasible. All three datasets are free and public: Disease Prediction Using Machine Learning (4,920 cases, 41 diseases, 132 symptoms), Chest X-Ray Images (Pneumonia) (5,863 images), and IU X-Ray / Open-i (3,955 reports). All tools are mature and covered in the track: Scikit-learn, TensorFlow/Keras, Hugging Face, Streamlit, MLflow, and Azure Machine Learning. Google Colab and Kaggle Notebooks provide a free GPU for the image model. We do not design new architectures; we use established backbones (ResNet/VGG) and pre-trained models, which keeps the technical risk low.
- **Operational Feasibility:** Feasible. The team has six members with clearly separated roles (Data, Symptoms ML, CNN, Transfer Learning, NLP, Website and Cloud), and every member can explain any part. The user flow is simple (three inputs, one summary page) and needs no training for patients. Work is coordinated through one GitHub repository, MLflow with an agreed naming convention, and a short weekly meeting.
- **Economic Feasibility:** Feasible. The project uses only free or student-tier resources (C-05): free datasets, free GPU notebooks, GitHub, open-source libraries, MLflow, and Azure student or free tier. No patient data is collected, so there are no data-acquisition costs. The main cost is team time over 4 weeks. The benefit is educational (applying the full track in one integrated project) and practical (a prototype that shows how patient guidance can reduce wrong-specialty visits).

Limitations That Affect Feasibility (stated openly):

- The pneumonia dataset is paediatric and has two classes only, so the model cannot detect other chest conditions and may not generalise to adults.
- The symptom dataset is clean and structured, so very high accuracy may not reflect real patients.
- The three datasets come from different patients, so the modules are independent and are not trained on a shared case.

2. Project Design

The Design section translates the verified analysis into architectural specifications, resource configurations, timelines, and execution strategies.

2.1 Project Objectives & Scope Constraints

Establishes definitive, measurable boundaries around project execution to ensure clear constraints.

SMART CLINIC: SCOPE DEFINITION

SMART Objective: Within 4 weeks of kick-off, deliver a published Smart Clinic website that accepts symptoms, a chest X-ray, and a report, and returns one downloadable summary within about 10 seconds. The three models should reach target accuracy on a held-out test set of about 90% or higher for symptoms and X-ray and about 85% or higher for report classification, reported honestly with Precision, Recall, and F1. All experiments must be logged in MLflow.

Measurable Objectives:

#	Objective	Measure	Target
O1	Predict possible diseases from symptoms	Accuracy, Precision, Recall, F1; top-3 output	About 90% or higher
O2	Classify chest X-ray as normal or pneumonia	Accuracy, Precision, Recall, F1, AUC	About 90% or higher

#	Objective	Measure	Target
O3	Classify the report and explain key terms	Accuracy, Precision, Recall, F1	About 85% or higher
O4	Understand Arabic and Egyptian symptom phrases	Sample phrases extracted correctly	Dictionary and Regex rules working
O5	One integrated summary with specialty, consistency check, and urgent warning	End-to-end test cases	All Must-have flows pass
O6	Publish online	Website reachable from desktop and mobile browsers	Live on Azure by Week 4

In-Scope:

- Chest and respiratory conditions.
- Symptom input in free text (English and Arabic/Egyptian phrases).
- Chest X-ray opinion (normal vs. pneumonia) with a confidence value.
- Report input as text or file; normal/abnormal result and term explanation.
- One downloadable summary page with a suggested specialty.
- Experiment tracking in MLflow, code on GitHub, deployment on Azure.

Out-of-Scope:

- Other medical specialties and other X-ray findings (for example tumours or fractures).
- Writing reports or giving a final diagnosis.
- Voice input, user accounts, medical history, appointment booking, and a mobile application.
- Collecting or storing real patient data.

Frozen Core: the project idea, the three inputs, the chest scope, one summary output, the medical disclaimer, and the delivery time. Other changes follow the change-control levels in the Requirements Gathering document.

2.2 Technical & System Architecture

Defines how the proposed solution is engineered and structured conceptually. The system is organised in layers: patient input flows through three independent models, then an integration layer, and ends in one summary page.

SMART CLINIC: DATA AND SYSTEM FLOW BLUEPRINT

Layer	What it does	Technologies
Data layer	Downloads, cleans, plots, and splits the three datasets (80% training, 10% testing, 10% validation) and saves clean copies in one shared place.	Python, Pandas, NumPy, Matplotlib, Seaborn
Input layer	Four pages: symptoms, X-ray upload, report, and final summary. Validates input and shows clear errors for wrong file types or empty text.	Streamlit
Arabic / extraction layer	Converts the patient's own phrases (Arabic, Egyptian dialect, or English) into the symptom names that Mind 1 understands, using a dictionary and Regex rules with basic text normalisation.	Python, Regex
Model layer: Mind 1 (Symptoms)	Predicts the closest three diseases with probabilities. Baseline Logistic Regression, then Decision Tree, Naive Bayes, KNN, SVM, Random Forest, and Boosting, with PCA / feature selection.	Scikit-learn
Model layer: Mind 2 (X-ray)	Classifies the image as normal or pneumonia with a confidence value. Compares a CNN from scratch with Transfer Learning (ResNet/VGG), using augmentation, class weighting, Dropout, and Early Stopping. Grad-CAM heatmap if time allows.	TensorFlow / Keras
Model layer: Mind 3 (Reports)	Cleans report text, converts it to TF-IDF, and classifies it as normal or needs follow-up, with the label taken from MeSH annotations. Compares with a Hugging Face model. Extracts key medical terms and explains them with a glossary.	NLTK, TF-IDF, Hugging Face
Integration layer	Maps predicted diseases to a suggested specialty, checks whether symptom, X-ray, and report results agree or differ, and raises an urgent-symptom warning.	Python, REST API
Presentation layer	One summary page with the disclaimer, a downloadable copy for the doctor, and a note about any missing input.	Streamlit
MLOps and cloud layer	Tracks every experiment, stores and serves the final models, and publishes the website.	MLflow, Azure Machine Learning, Azure deployment, GitHub

Data and System Flow:

1. The patient types symptoms, uploads a chest X-ray, and types or uploads the report.
2. The Arabic layer extracts symptom names from the text and shows them back for confirmation.
3. Mind 1 returns the top three possible diseases; Mind 2 returns the image opinion and confidence; Mind 3 returns the report result and explained terms.
4. The integration layer adds the suggested specialty, the consistency check, and any urgent warning.
5. The summary page is shown with the disclaimer and can be downloaded.

Inputs are processed for the current session only and are not stored permanently. If one input is missing, the page shows the available results and states which input is missing.

Design Principles: the three models are independent so the team can work in parallel; every result is worded as guidance, not diagnosis; and every experiment is logged so that model choices can be justified.

2.3 Timeline & Risk Matrix

Outlines the tactical roadmap alongside managed risk factors identified during system design.

SMART CLINIC: EXECUTION SCHEDULE & RISKS

Phased Roadmap (4 weeks):

Phase	Focus	Main activities	Key deliverables
Week 1	Setup, data, baselines	Kick-off meeting; accounts (Kaggle, GitHub, Azure); project structure and MLflow set up before any training; Member 1 downloads, checks, plots, and splits the data; Members 2 to 5 run a first simple model.	GitHub repository and README; MLflow running; clean split datasets and plots report; baseline model per part
Week 2	Model development and comparison	Symptom models compared with error analysis; CNN improved; Transfer Learning built and compared; TF-IDF classifier and Hugging Face tested; website pages designed and Streamlit skeleton prepared.	Comparison tables; all experiments in MLflow
Week 3	Best models, Arabic layer, website	Random Forest and Boosting; best model chosen for each part; Arabic dictionary and Regex rules; website pages built with disclaimer; test cases prepared; wireframes reviewed with the Project Owner.	Best saved model per part; Arabic dictionary; website running locally
Week 4	Integration, deployment, delivery	Models connected to the website; one downloadable summary; models uploaded to Azure ML and website published; end-to-end testing; final report, presentation, and live demo rehearsal.	Website online; final report, presentation, live demo; organised GitHub code

Risk Matrix:

Identified Risk Event	Probability	Impact	Designed Mitigation Strategy
Output is misread as a diagnosis	Medium	High	Disclaimer on every page; wording such as “leans towards”; clear statement that it is not a substitute for a doctor.
Class imbalance and a very small original validation split in the X-ray data	High	Medium	Stratified re-split, data augmentation, class weighting, and reporting Precision, Recall, F1, and AUC, not accuracy alone.
X-ray model does not generalise (paediatric, two-class dataset)	High	Medium	State the limitation openly; show confidence carefully; do not claim detection of other conditions.

Identified Risk Event	Probability	Impact	Designed Mitigation Strategy
Overstated accuracy on the clean symptom dataset	Medium	Medium	Analyse confused diseases, report all metrics, and avoid presenting results as real-world performance.
Arabic or Egyptian phrases not covered by the dictionary	Medium	Medium	Add common Egyptian expressions, basic normalisation, and test with sample phrases; show extracted symptoms back to the patient for confirmation.
Integration delays between the three models and the website	Medium	High	Agree on input/output formats in Week 1; Member 6 builds a skeleton early; Members 2 to 5 support integration in Week 4.
Free GPU or Azure free-tier limits slow training or deployment	Medium	Medium	Use Colab and Kaggle Notebooks as alternatives; save checkpoints; start Azure setup early; keep models lightweight.
Urgent symptoms not flagged	Low	High	Add an urgent-symptom rule list and a clear warning to seek immediate care; test with sample cases.
Team time shortage within 4 weeks	Medium	High	Follow MoSCoW priorities; the heatmap and AraBERT are Could-have items and are dropped first; weekly progress meeting.
Dataset licence or availability issue	Low	Medium	Keep local clean copies; cite sources; replace a dataset only through the change-control process.

Next Step: this document, together with the Requirements Gathering baseline, guides the build in Weeks 1 to 4. Any change that affects scope, cloud provider, or a Must-have requirement follows the change-control process and needs Project Owner approval.